From Playbook to Production: Scoring Big with Gemma 4

Unlocking the power of Google's flagship open models

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About Me

- **Staff Data Engineer, FanDuel:** Architecting enterprise-scale data infrastructure, semantic layer, and building secure, high-throughput AI pipelines.
- **Google Expert (Cloud):** Recognized by Google for expertise in Google Cloud Platform, GenAI, and Vertex AI. Active public speaker.
- **Academic Background:** Ph.D. in Analytics & Data Science (KSU) | MPH in Global Epidemiology (Emory Univ.).
- **Previous Career Milestones:**
  - *Intuit Mailchimp:* Analytics Enablement guru.
  - *CDC:* 10 years as an Epidemiologist & Biostatistician.
  - *Mirry.AI:* Former Chief AI Officer.

Connect with Me

- **LinkedIn:** linkedin.com/in/jmrudd
- **GitHub:** github.com/JessicaRudd
- **Substack:** funsizedatabytes.substack.com
- </> **GDE Profile:** g.dev/jmrudd

Gemma vs. Gemini: Sibling Models, Different Missions

Feature	Gemma (Open Weights)	Gemini (Proprietary API)
Hosting	Runs inside your subnets (GKE, VMs, or offline)	Google-managed infrastructure (Vertex AI)
Network Boundary	100% Zero-Egress (stays in VPC subnet)	Access via VPC Service Controls
Customization	Full parameter access; fine-tune via QLoRA	Prompting, system instructions
Max Context	Up to 256K tokens	Up to 2M+ tokens
Cost Model	Pay-for-Compute (flat infra cost)	Pay-per-Token (variable API billing)

Why Choose Gemma? Core Strengths

Strict Data Privacy & Compliance

Ideal for: Healthcare, finance, and legal sectors. No data ever leaves your secure boundary.

Low Latency & Zero Dependency

Ideal for: On-device applications, mobile terminals, and offline field operations.

Domain-Specific Customization

Ideal for: Fine-tuning on proprietary codebases, docs, or specialized database schemas.

Cost Predictability at Scale

Ideal for: High-throughput automated pipelines. Pay a flat rate for GPU/TPU compute.

The Gemma 4 Spectrum: A Model for Every Mission

E2B & E4B
Edge & Mobile

VRAM: < 4GB

Use Case: On-device execution. Fits perfectly on smartphones and IoT devices for offline chat and translation.

12B Unified
The Dev Sweet Spot

VRAM: ~8-12GB

Use Case: The ultimate local companion. Runs flawlessly on a standard MacBook Pro for daily coding and API tasks.

26B MoE
High Throughput

VRAM: ~16GB

Use Case: Mixture-of-Experts. Lightning fast inference matching 31B intelligence, but only using 4B active parameters per token.

31B Dense
Heavy Duty

VRAM: 24GB+

Use Case: Flagship server reasoning. Built for autonomous agentic workflows, complex math, and enterprise SQL generation.

Under the Hood: Architectural Highlights

256K Context Window

128K context for Edge (E2B/E4B), scaling up to 256K for Mid/Large models.

Native Multimodality

Text, image, and native audio processing directly in the network (no separate transcoder).

Native Thinking Modes

Configurable reasoning capabilities using <|think|> tags to plan before responding.

Multi-Token Prediction

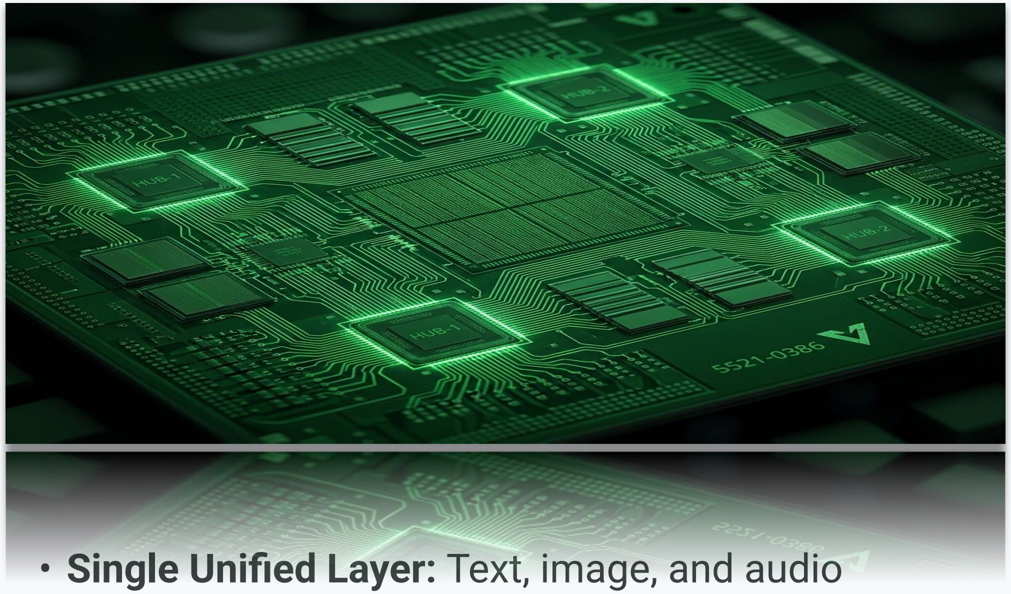
Predicts multiple tokens simultaneously, leading to a 2x-3x speedup in inference generation.

Architectural Shift: Removing Encoder Bloat

The Old Paradigm

- **Separate Encoders:** Dedicated vision & audio models duct-taped to a text core.
- **Heavy Parameter Bloat:** Massive size increases.
- **High Latency:** Slow, multi-step processing.

Gemma 4 Innovation



- **Single Unified Layer:** Text, image, and audio processed natively.
- **Lean & Efficient:** Drastically reduced parameter count.
- **Low Latency:** Fast, one-step reasoning.

The Local Advantage: Doing More with Less

By stripping away the baggage of multiple encoders, Gemma 4 achieves something remarkable for the open-weights ecosystem.

- **Lower Parameter Count:** Without separate vision and audio blocks, the total model footprint is dramatically reduced.
- **Smoother Local Execution:** Models like the 12B variant can now run fluidly on standard developer laptops without thrashing system memory.
- **Native Multimodality at the Edge:** You get true out-of-the-box text, vision, and audio capabilities without needing enterprise-grade GPU clusters.

Deployment Flexibility

Local Execution

Tools: Ollama, vLLM, Docker.

Ideal for: E2B, E4B, 12B. Private offline testing, no data leaves your laptop.

Google AI Studio

Tools: Developer-first API, playground.

Ideal for: Rapid prototyping, low-friction serverless integration, and testing thinking modes.

Google Cloud Production

Tools: Model Garden, Vertex AI Endpoints, GKE, Cloud Run.

Hardware: TPUs (v5/v6e) and GPUs. Fully managed, auto-scaling MLOps.

Local Deployment: Installing Ollama

It takes less than 3 minutes to get Gemma 4 running securely on any machine.

MacOS

Install effortlessly via Homebrew.

```
brew install --cask ollama
```

Linux

Download and execute the official script.

```
curl -fsSL
https://ollama.com/install.sh |
sh
```

Windows

Download the .exe or use WSL2 (Recommended).

```
# Inside WSL2 terminal:
curl -fsSL
https://ollama.com/install.sh |
sh
```

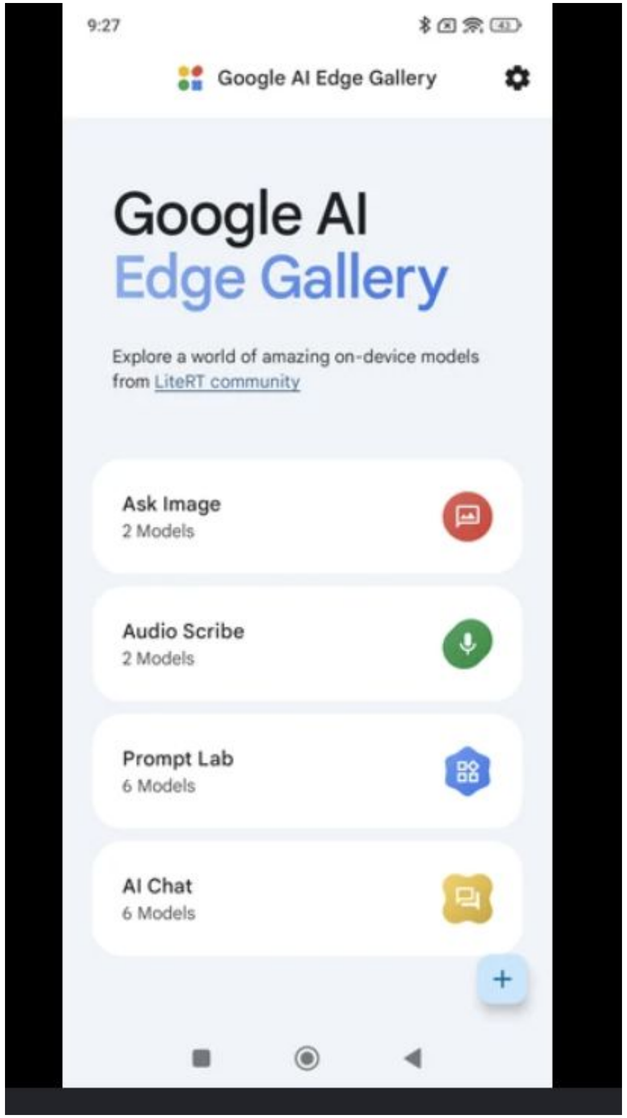
Unified Run Command: Once installed on any OS, simply type `ollama run gemma4:12b` to download the weights and start your local server.

Real-World Edge: Google Edge Gallery App

Taking Gemma offline for travel

Next month, I'm traveling to Argentina. I won't always have reliable cell service, and roaming is expensive. By running Gemma E2B locally on my phone via the Edge Gallery app, I have a powerful offline companion.

- **Offline Translation:** Instant English-to-Spanish translation powered entirely on-device.
- **Local Context Queries:** Asking questions about pre-downloaded itineraries without hitting an external server.
- **Privacy First:** No hotel Wi-Fi snooping; all inference and data stays strictly on my device.



Demo 1: Visual UIs for Local LLMs

Want a "ChatGPT-like" experience running completely offline? You have two great options.

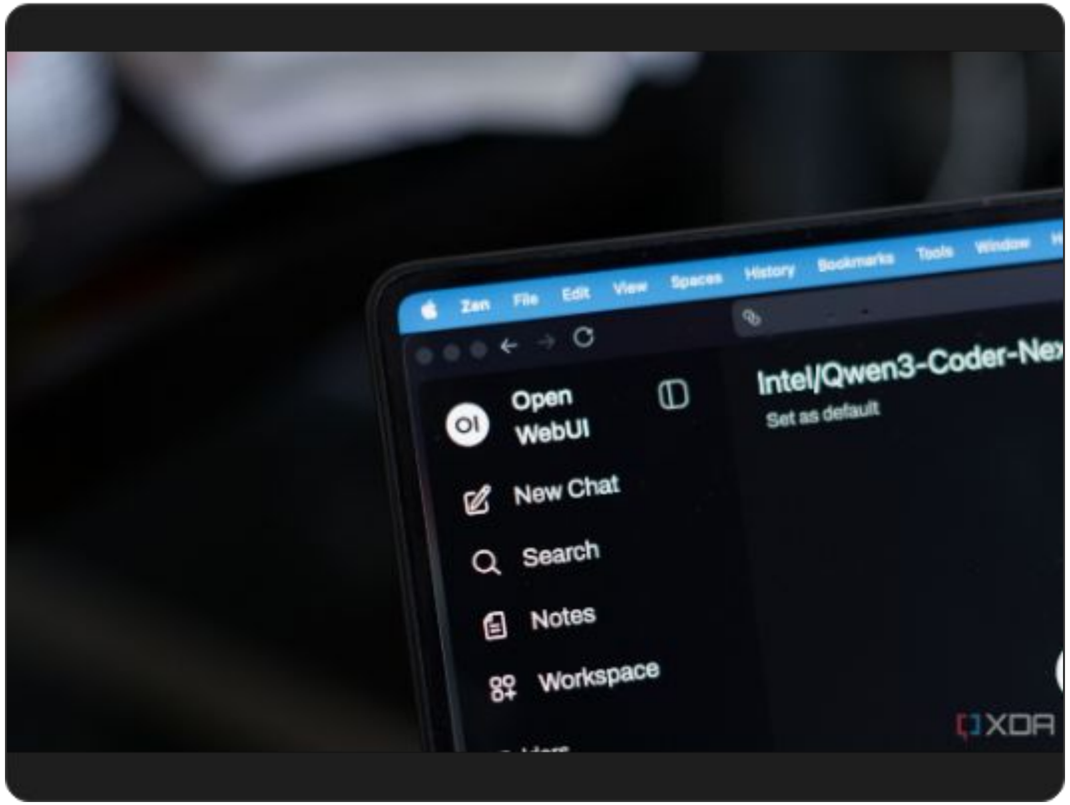
Open WebUI

A sleek web interface that connects directly to Ollama. Runs via Docker and features web search, RAG, and multi-model chat.

```
docker run -d -p 3000:8080 --add-host=host.docker.internal:host-gateway -v open-webui:/app/backend/data ghcr.io/open-webui/open-webui:main
```

LM Studio

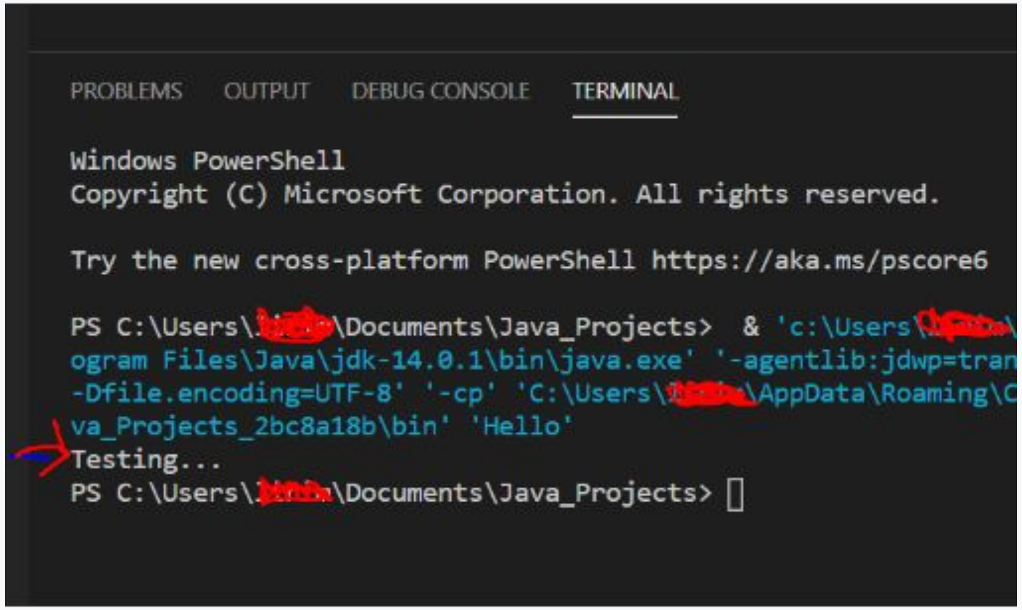
A powerful desktop application (Mac/Win/Linux). Perfect for discovering new GGUF models visually, downloading them, and running a built-in local inference server.



Demo 2: Terminal & IDE Integration

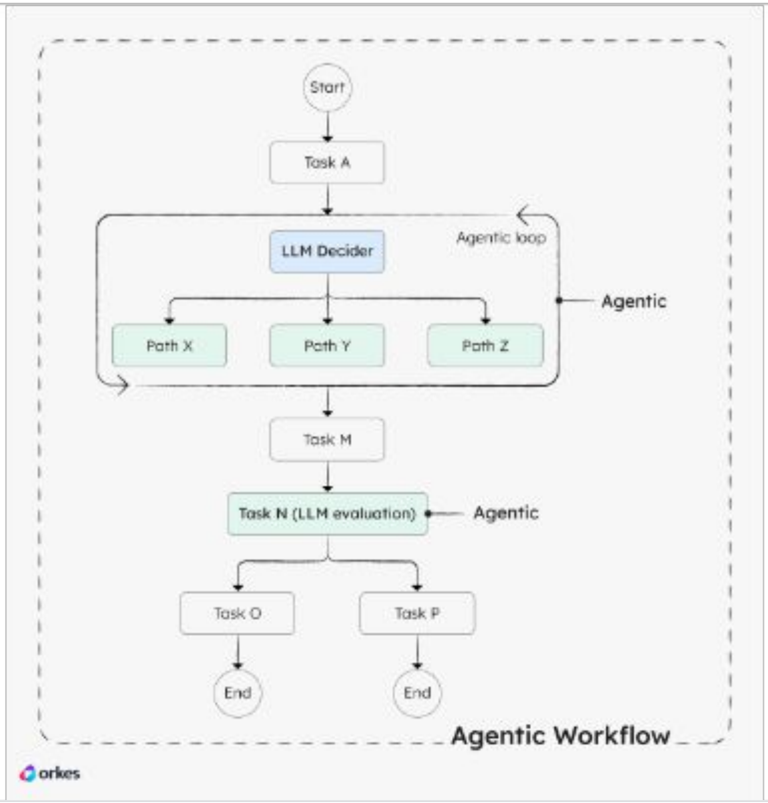
For developers, the GUI isn't always necessary. Ollama exposes a local API at localhost:11434 by default.

- **Terminal Chat:** Instantly query the model directly in your command line for quick, distraction-free answers without switching context.
- **Local IDE Integration:** Connect VS Code extensions (like Continue.dev) to your local endpoint for private, offline code completion and refactoring.
- **Python Automation:** Write scripts to parse JSON, summarize server logs, or clean data—all without incurring any cloud API costs.



The Shift to Agentic Workflows

- **Traditional LLM:** Text in → Text out.
- **Agentic Gemma 4:**
  Text/Media in → Plan → Call Tools → Structured Action.
- **Gemma-Skills & ADK:**
  - Standardized directory structures for agent tools.
  - Native integration with MCP (Model Context Protocol).
  - Secure tool execution sandboxes (e.g., Cloud Run Proxy).



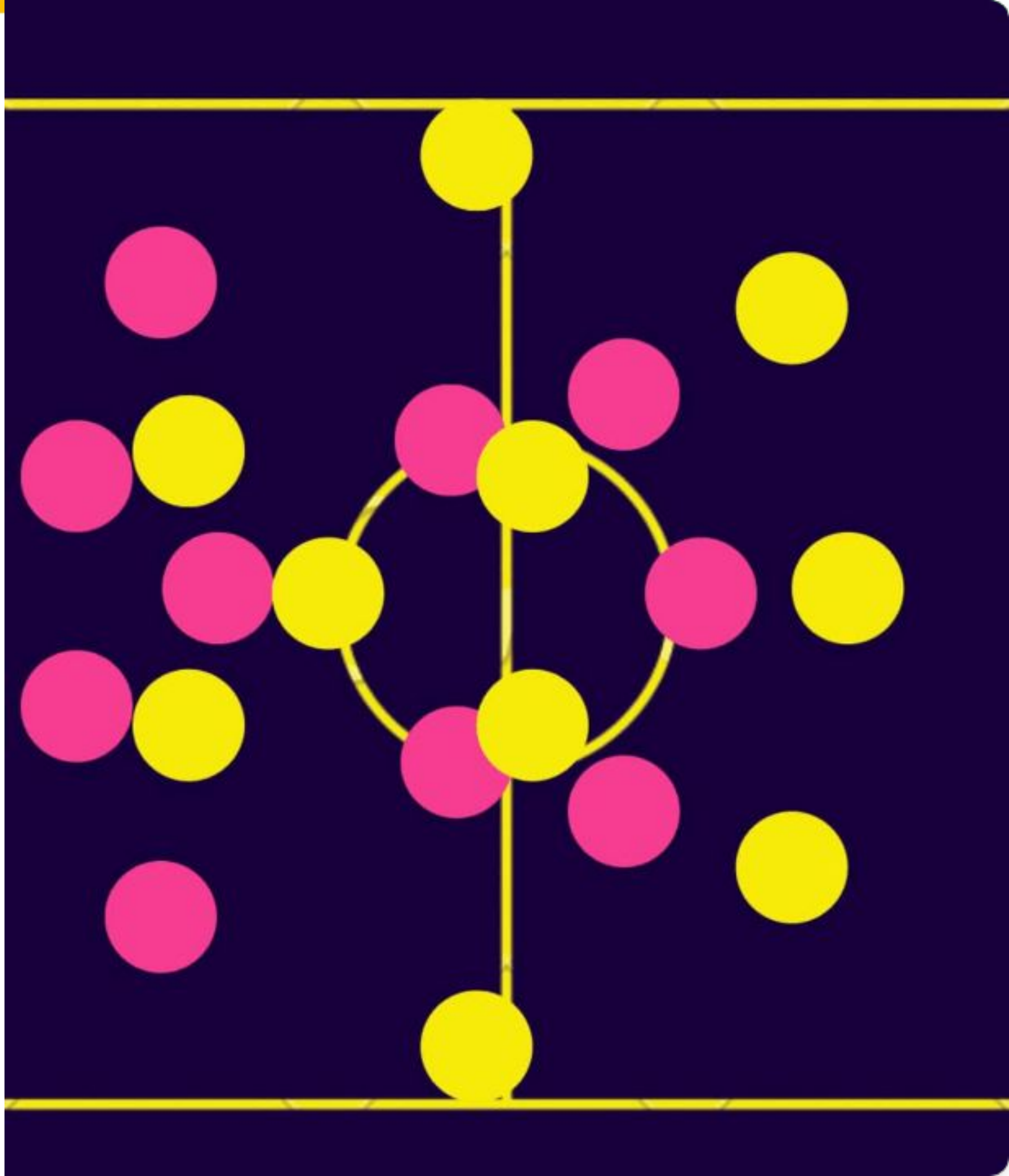
Demo 3:
The World Cup Analyst

Highlighting Gemma 4's Advanced Capabilities

The Scenario: A high-stakes match situation (e.g., 90th minute penalty, Canada vs USA).

Core Tasks Demonstrated:

- **Thinking Mode:** Forcing the model to analyze tactics, history, and psychological pressure *before* generating commentary.
- **Structured JSON Output:** Guaranteeing the response fits a strict schema for downstream app ingestion.
- **Real-Time Generation:** Fast token output via Python script execution.

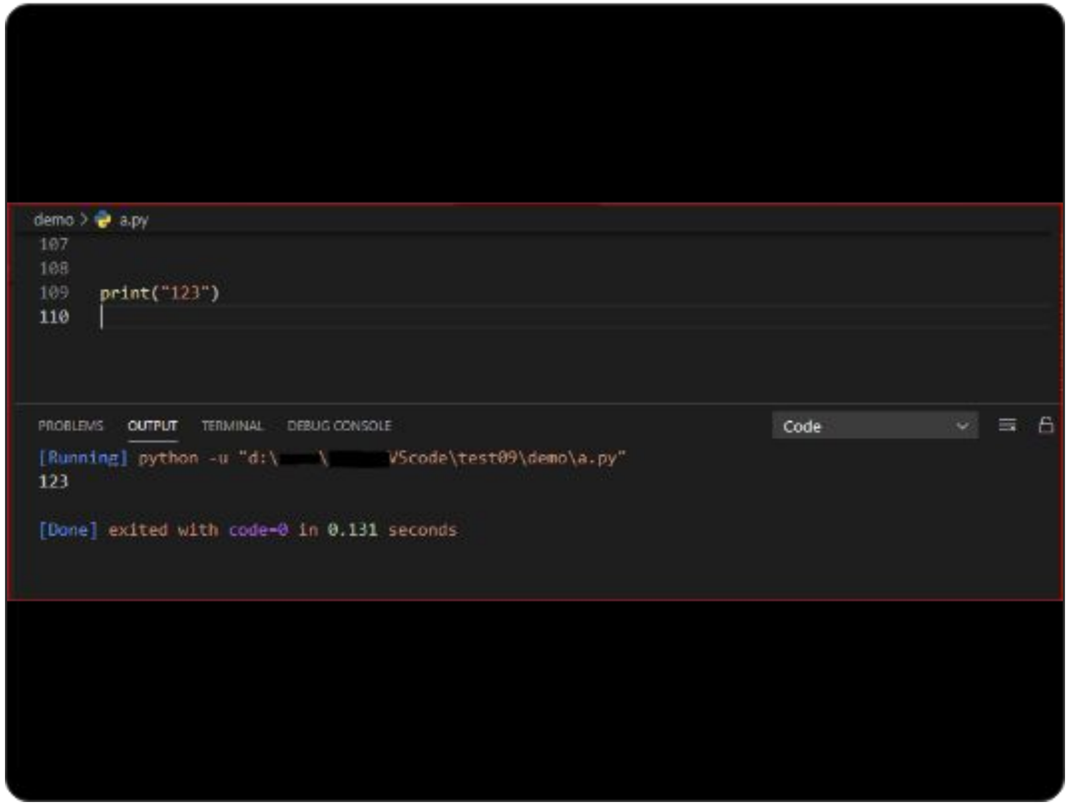


Live Execution

```
# Option A: Cloud Mode (AI Studio)
export GEMINI_API_KEY="your-api-key"
python3 world_cup_analyst.py

# Option B: Local Mode (Ollama Backup)
# Ensure 'gemma4:12b' or 'gemma4:4b' is running
python3 world_cup_analyst.py --local
```

Switching to live terminal...



Resources & Next Steps

- **Demo Code & Slides:** https://github.com/JessicaRudd/gemma4-demo
- **Read My Newsletter:** funsizedatabytes.substack.com
- **Gemma 4 Docs:** ai.google.dev/gemma
- **LinkedIn:** linkedin.com/in/jmrudd

Thank You! Q&A